

Alexey Morozov

Backend Engineer (Python / Go)

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Remote or hybrid English — B2

Professional Summary

Backend engineer with 5 years of experience, 3 of them in fintech. I build and run microservices in Python (FastAPI, asyncio) and Go: the payment gateway my team owns handles around 400 requests per second at peak. Particular strength in PostgreSQL beyond CRUD — query plans, indexing and partitioning. I take services through to production rather than to merge: tests, metrics, and on-call incident analysis.

Technical Skills

Languages: Python (5 years, asyncio, typing), Go (2 years, services in production), SQL, Bash, Git; Rust — basic

Frameworks and protocols: FastAPI, aiohttp, Celery, gRPC, REST API; Django — 1.5 years

Databases and queues: PostgreSQL (query plans, indexing, partitioning), Redis (caching, distributed locks); Kafka — producing and consuming in production tasks; MongoDB — basic

Infrastructure: Docker, GitLab CI/CD (Continuous Integration/Delivery), Linux; Kubernetes — deploying via existing manifests, debugging pod failures

Architecture and reliability: microservice decomposition, idempotency, retries, partial-failure handling; saga pattern — limited exposure

Observability and testing: Prometheus, Grafana, Sentry, structured logging, pytest, testcontainers

Experience

Aurora Fintech

March 2023 — Present

Backend Engineer

- Build and maintain a payment gateway in Python (FastAPI, asyncio): payment intake, provider reconciliation, retries and operation idempotency.
- Designed the idempotency scheme for payment operations, eliminating double charges on retried requests.
- Ported two internal services from Python to Go, roughly halving response time on heavy endpoints.
- Tuned PostgreSQL: query plans, missing indexes, monthly partitioning of the transactions table.
- Set up Prometheus and Grafana metrics and alerts for my services; on-call rotation, incident analysis and postmortems.
- Integration and unit testing with pytest and testcontainers across critical paths.
- Code review and mentoring: two interns, both hired after their internship.

Cityline Retail

July 2021 — February 2023

Backend Engineer

- Developed catalogue and cart services for an e-commerce platform in Python (Django, later FastAPI), exposing REST APIs to web and mobile clients.
- Moved price-list export to asynchronous tasks (Celery, Redis), unblocking the main request path and fitting the nightly window.
- Built a Redis caching layer for the catalogue, cutting peak load on the primary database.
- Extracted the catalogue service during a monolith split, including data migration.
- Maintained GitLab CI/CD pipelines: tests, linters, image builds.

Cityline Retail

February 2021 — June 2021

Software Engineering Intern

- Fixed bugs in an existing codebase, wrote unit tests, learned to navigate a large legacy code base. Received a full-time offer after the internship.

Projects

Rate limiter in Go

personal, open source

- Rate-limiting library: token bucket and sliding window, Redis-backed state for distributed mode, benchmarks and race-condition tests.
- Adopted by one of the internal services at work.

Database schema migration tool

internal library at work

- CLI on top of Alembic: ordered migration runs across several services, rollback, and pre-deploy checks for incompatible changes.
- Extracted into an internal library, used by three teams.

Education

Ural Technical University — B.Sc. Software Engineering (2017–2021)

Additional training

- Go for backend engineers — ported two internal services from Python to Go as the practical outcome.
- PostgreSQL query plans and optimisation — slow query log analysis, query rewriting, index selection.

Awards

- Fintech Hackathon 2022 — 2nd place, team category.